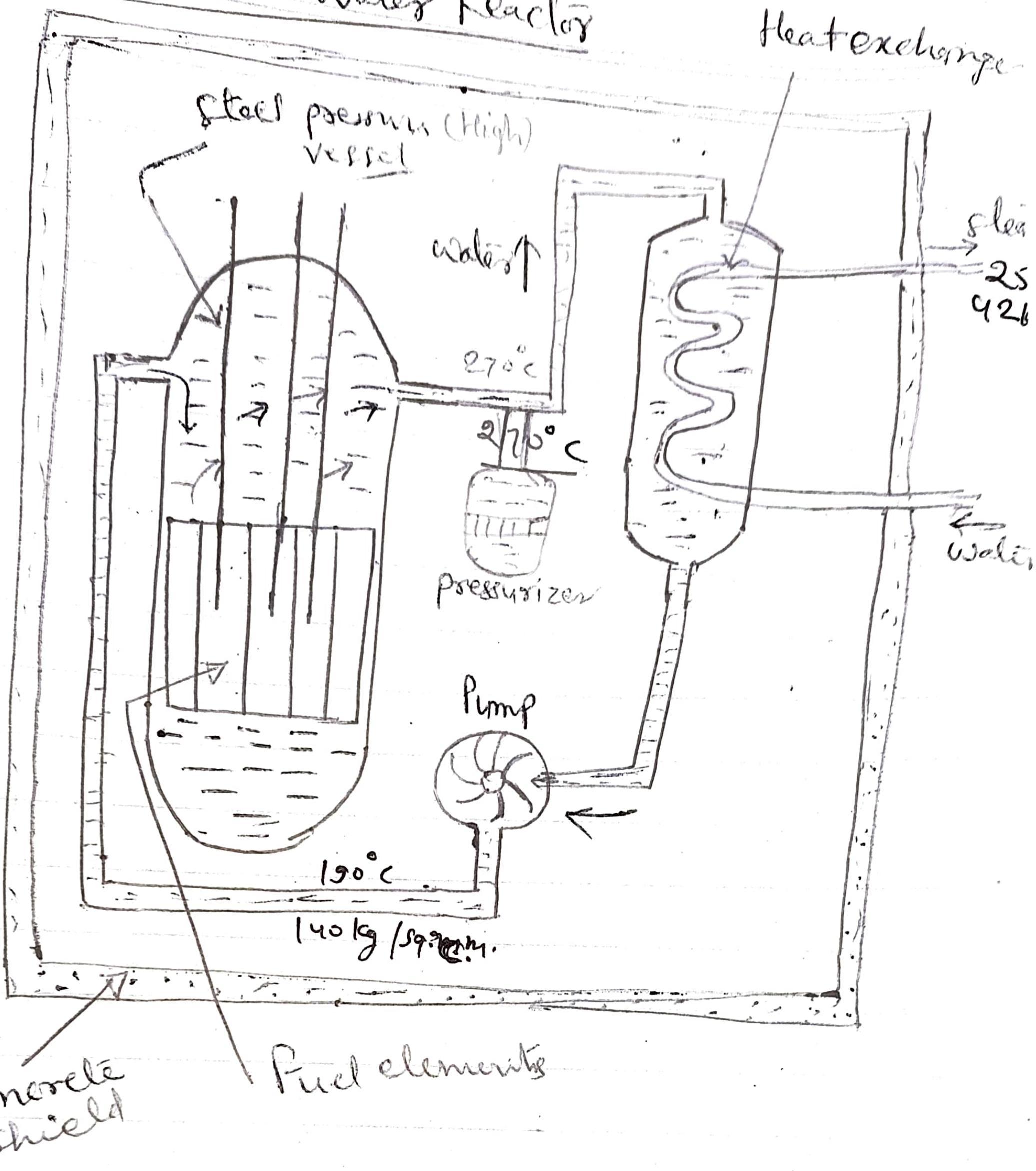
(PWR)

Pressurised Water Reactor




Fuel used in PWR is enriched Uranium oxide ($UO_2$) (3% $U^{235}$)

Pressurised water is used both as coolant and moderator. Water is passed in to reactor at 190°c and 140 kg/sq.cm. (2000 psi) pressure and is discharged from reactor at 270°c. This water is passed through heat exchanger where steam is raised. Steam is of rather poor quality

Steam
Temp – 270°C, pressure 42 kg/cm²
Steam is condensed in the condenser and condensate returns to heat exchanger.
drawback → Requires high strength pressure vessel
advantage → Steam supplied to the turbine is completely free from contamination
overall efficiency – 33%

Rajasthan Atomic Power station, Madras Atomic station and Narora Atomic Power project are of PWR type.

However they use heavy water as coolant and moderator. (PHWR → Pressurised heavy water Reactor)

* The Canadian Candu reactor is thermodynamically a pressurised water reactor though it is a bit diff. mechanically, particularly in the subdivision of the pressure circuit. It uses natural Uranium as fuel and Heavy water (280/kg) as coolant and Moderator.

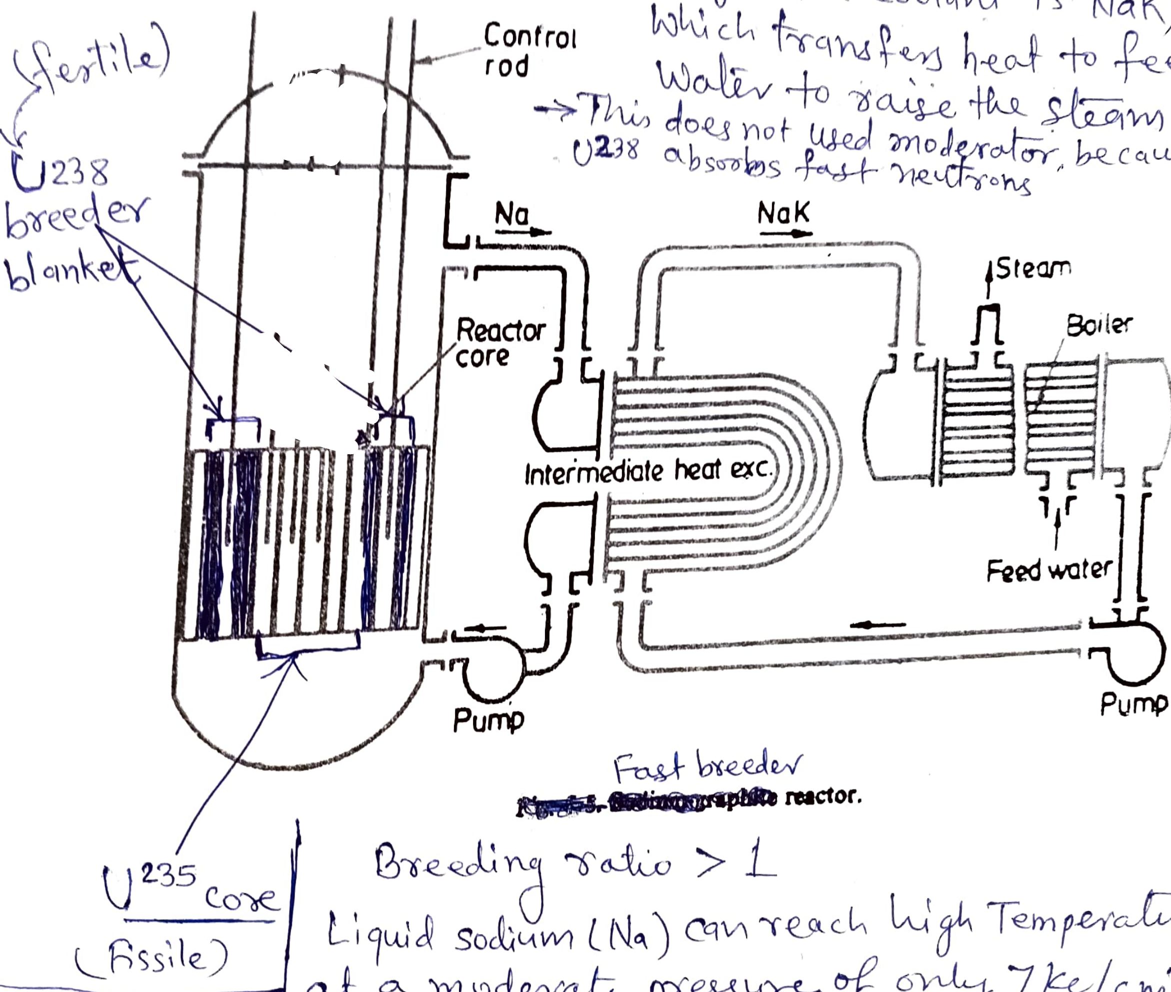
It was first developed by canadians so called CANDU (Canadians Deuterium Uranium)

* CANDU – uses Natural Uranium ($U^{238}$) as a fuel and Heavy water as a Coolant and Moderator.

Deuterium oxide called heavy water used as moderator

**Fast Breeder Reactor** (FBR): It produces heat and at the same time converts fertile material into fissile material. In this reactor the core containing $U^{235}$ is surrounded by a blanket of fertile material i.e. $U^{238}$ which gets converted into $Pu^{239}$ (fissile material). Two heat exchangers are used. The reactor core is cooled by liquid sodium (Na). In the secondary heat exchanger the coolant is NaK, which transfers heat to feed water to raise the steam.

→ This does not used moderator, because $U^{238}$ absorbs fast neutrons.




Fast breeder reactor.

Breeding ratio > 1

Liquid sodium (Na) can reach high Temperature at a moderate pressure of only $7\ kg/cm^2$
so, Intermediate heat exchanger is necessary between reactor and the boiler.

Coolant leaves reactor at 640°c and steam temperature is 540°c.

FBR also uses 20% $PuO_2$ + 8% $UO_2$ as fuel and Thorium as ~~fuel~~ fertile material which converted in to $U^{233}$ which is fissionable. [worlds largest deposits of thorium about 4,50,000 tones in the form of sand found in Kerala and Gopalpur coast of orrisa.

Typical power densities (MW/$m^3$) in fission reactors cores are

| | |
|---|---|
| Gas cooled ~~co~~ reactor | : 0.53 MW/$m^3$ |
| High Temp. gas cooled " | : 7.75 MW/$m^3$ |
| Heavy water " | : 18.0 MW/$m^3$ |
| Boiling water " | : 29.0 " " |
| Pressurrised water " | : 54.75 " " |
| Fast Breeder reactor | : 760 MW/$m^3$ |